Supplementary Material

Whose Writing Is This, Anyway?: Lessons from Building a Free and Open Citation Hallucination Detector

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This document is the supplementary material for the article named above, published in *The Code4Lib Journal*. It collects the statistical and algorithmic detail that the article deliberately keeps out of its running text: Supplement A gives the formulas, thresholds, and confidence intervals behind the reported numbers, and Supplement B gives the full validation procedure, including request pacing, retry policy, and the rules governing each status.

It is archived with the code at `https://doi.org/10.5281/zenodo.21795712` and tracked at `https://github.com/OhioMathTeacher/citation-validator-app`.

A Statistical and algorithmic details

The article deliberately keeps formulas and parameter values out of its running text, so that an editor or scholarly-communications reader can follow the argument without wading through statistical or algorithmic minutiae. This supplement collects that material in one place for readers who want to inspect, reproduce, or modify the calculations.

A.1 Title-matching: Jaccard similarity

Title similarity between a BibTeX entry and a database record is computed using word-level Jaccard similarity (Jaccard 1912):

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{1}$$

where A and B are the sets of lowercase alphanumeric tokens extracted from the two titles. A value of 1 means identical token sets; 0 means no overlap.

A.2 Match thresholds

Citation Validator applies the following thresholds, established by spot-checking on the curated test sets:

- **Step 1 (DOI resolution).** A Jaccard score below 0.4 between the BibTeX title and the title CrossRef returns for the same DOI triggers a title-mismatch warning and potential escalation to *suspicious*.
- **Step 2 (OpenAlex).** A Jaccard similarity ≥ 0.5 between the query title and the top OpenAlex result constitutes a strong match; ≥ 0.3 is noted but not validating.
- **Step 2 (Semantic Scholar).** A strong match requires both title similarity ≥ 0.5 *and* author/year cross-validation.

- **AI second pass.** The model returns a structured JSON response with an `is_suspicious` flag and a confidence score from 0–100. Confidence ≥ 70 with `is_suspicious: true` escalates the citation to *suspicious*; confidence ≥ 85 escalates to *invalid*.

A.3 Confidence intervals: Wilson score

Confidence intervals on proportions (e.g. false-positive rate, recall) are computed using the Wilson score interval (Wilson 1927), which provides reliable coverage even for proportions near 0 or 1 with small sample sizes. For k successes in n trials, the $1-\alpha$ Wilson interval is:

$$\frac{1}{1+z^2/n}\left(\hat{p}+\frac{z^2}{2n}\pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}\right) \tag{2}$$

where $\hat{p} = k/n$ and $z = z_{1-\alpha/2}$. All intervals reported in the article use $\alpha = 0.05$ (95% confidence). The Wilson interval is preferred over the more familiar normal-approximation ("Wald") interval because the Wald interval is degenerate at $\hat{p} = 0$ or $\hat{p} = 1$ and gives poor coverage for small n.

A.4 Wilson intervals for the headline numbers

The following table collects the Wilson 95% intervals for the proportions reported in the article. Readers who want a sharper sense of uncertainty around any single number will find it here.

- Deterministic FPR, 0/391 real citations: [0.0%, 0.96%]
- Gemini recall on Ansari fabrications (92/100): [85.0%, 95.9%]
- Claude recall on Ansari fabrications (85/100): [76.7%, 90.7%]
- Gemini FPR on arXiv preprints (146/285): [45.4%, 57.0%]

A.5 Detection metrics: precision, recall, and FPR

Following standard detection convention, the four outcome counts are:

- A *true positive* is a fabricated citation correctly flagged.
- A *false positive* is a real citation incorrectly flagged.
- A *true negative* is a real citation correctly left alone (status *valid* or *warning*).
- A *false negative* is a fabricated citation that slipped through unflagged.

These four counts let us quantify three metrics of Citation Validator's accuracy: *precision*, *recall*, and the *false-positive rate (FPR)*.

- Precision asks: of the citations Citation Validator flagged, what fraction were actually fabricated?
- Recall asks: of the fabricated citations in the test set, what fraction did Citation Validator catch?
- FPR asks: of the real citations, what fraction did Citation Validator wrongly flag?

Precision and recall both reward catching fabrications; FPR penalizes the failure mode that matters most in scholarly settings, wrongly flagging a real citation. The aggregate metrics are computed from those counts as follows: The terms *true positive*, *false positive*, *true negative*, and *false negative* are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

B Validation procedure in detail

This supplement gives the full algorithmic detail behind the citation-validation procedure. The article kept this material out of its running text so that an editor or scholarly-communications reader could follow the argument without wading through Citation Validator's request behavior, fallback policy, and matching thresholds. Readers who have opted in to the mechanism will find what they need here. The procedure is summarized in Figure 1 of the article; the prose that follows fills in the request behavior, retry policy, and thresholds the diagram leaves out.

B.1 Step 1: DOI resolution

A DOI—a Digital Object Identifier—is a permanent link assigned to a published work. When a citation includes one, it is the most reliable piece of metadata available: legitimate DOIs do not expire, and each one points to exactly one paper. Step 1 asks, in effect, *does this DOI lead where the citation claims it does?*

The deterministic code first queries CrossRef, the largest DOI registry. CrossRef returns the metadata it has on record for that DOI: the title, authors, and year. Three outcomes are possible:

- **The DOI does not resolve.** CrossRef has no record of the DOI at all—the digital equivalent of a phone number that does not exist. This is strong evidence of fabrication, and the citation is classified as *invalid*.
- **The DOI resolves and the returned metadata matches the citation.** The title, authors, and year on record line up with what the BibTeX entry claims. The citation is classified as *valid*.
- **The DOI resolves but the returned metadata does not match.** The DOI is real, but it points to a different publication than the one the citation claims. This is the *stolen DOI* pattern: a real identifier attached to a fabricated citation. The citation is classified as *suspicious*.

Backup services. If CrossRef cannot be reached or returns an error, Citation Validator falls back to two further DOI-resolution services in sequence before giving up. Citations whose DOIs follow the arXiv format are routed to arXiv's own metadata service instead, because arXiv maintains its own records for its preprints.

Pacing requests. All of the services Citation Validator queries are free and public. Each places a cap on how often it can be called—typically a small number of requests per second. Citation Validator paces its requests so that it stays well under each service's cap (arXiv, the strictest, is queried at most once every three seconds). If a request fails because of a temporary problem—a server timeout, a refusal because the cap was hit, a transient network error—Citation Validator waits and tries again, up to four attempts spaced one, two, and four seconds apart. A definitive refusal (such as a "not found" response) is final and does not trigger a retry.

When a lookup still fails. A citation whose DOI cannot be resolved after all four attempts is treated as *unverified* rather than fabricated: it stays at status *warning* and is never escalated to *invalid* on the strength of a failed lookup. The distinction is central to the false-positive rates reported in the Results.

Title comparison. When a DOI resolves successfully, the title returned by the service is compared to the title in the BibTeX entry using a word-overlap measure (formula in Supplement A). If the two titles share most of their words, the citation passes through to *valid*; if they share few or none, the title-mismatch is taken as further evidence of fabrication, and the citation may be escalated to *suspicious*. Author matching is similarly tolerant: at least one last name from the BibTeX `author` field must appear somewhere in the resolved author list, ignoring differences in capitalization. Year matching requires an exact match.

B.2 Step 2: Database search by title

When a citation has no DOI, Citation Validator has nothing to resolve. Instead, it searches two open scholarly databases by title:

- **OpenAlex** (OpenAlex 2026): the title is sent as a search query, and the top result is inspected. If its title overlaps strongly with the citation's title (thresholds in Supplement A), the match is treated as validating; weaker overlaps are noted but do not validate.
- **Semantic Scholar** (Semantic Scholar 2026): a free-text title search returns title, authors, year, venue, and any other identifiers on file. A strong match here requires both a high title-overlap score *and* agreement on author and year.

If neither database returns a match, the citation is classified as *warning*—**not** *suspicious*. This is the critical design decision underlying everything else in the paper. Many legitimate citations are absent from these databases: arXiv preprints, non-English journals, older publications, books, and entire areas of scholarship that have not been indexed. Treating absence as evidence of fabrication would produce unacceptable false-positive rates on exactly the kinds of citations a small-journal editor most often encounters, as the experiments below confirm.

B.3 The optional AI second pass

For users who want a second opinion, citations with status *warning* or *suspicious* can be routed to a large language model for analysis. Citation Validator supports four providers:

- Gemini 2.5 Flash (Google; free tier, 1 million tokens per day).
- GPT-4o (OpenAI; paid).
- Claude Sonnet 4 (Anthropic; paid).
- Llama 3.3 70B (Meta, served via Groq; free tier, rate-limited).

The model receives the citation's BibTeX fields (entry type, title, authors, year, journal or booktitle, DOI) along with any metadata the deterministic code retrieved during Step 1 or Step 2. It is then asked to assess whether the citation shows signs of fabrication—internal inconsistency, the *Frankenstein* pattern of fragments stitched from multiple real papers, or metadata that conflicts with what the deterministic code already verified.

The model returns a structured response with a flag and a confidence score; the confidence thresholds for escalation to *suspicious* or *invalid* are listed in Supplement A. The AI second pass is not enabled by default. The deterministic code is the recommended procedure for everyday use; the rationale for this recommendation is developed in the Results.

References

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